

Dehua Lai

ML Infrastructure Engineer

Great Seattle Area

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SUMMARY

ML infrastructure engineer with 14+ years of experience building production machine-learning platforms and services. Deep experience in feature stores, online inference, data pipelines, platform reliability, and ML observability, including work on Uber Michelangelo and Oracle Cloud Infrastructure.

EXPERIENCE

Oracle Cloud Infrastructure (OCI) | Principal Member of Technical Staff

FEB 2026 - Present

- Led end-to-end development of an AI Cost Control platform that turns raw AI workload telemetry into cost, usage, and intent-level insights.
- Designed a raw-data ingestion pipeline for distributed AI request, usage, and billing sources; built Spark jobs to validate, enrich, and transform high-volume usage events into analytics-ready datasets.
- Built inference-serving infrastructure for the AICC platform, hosting Smart Model Router to predict request intent and recommend appropriate models.
- Developed aggregation pipelines for cost and usage metrics across key dimensions, supporting granular analysis of AI workload consumption.
- Led delivery of Langfuse-based LLM observability for OCI applications, including tracing and telemetry for prompts, responses, latency, token usage, execution paths, evaluation, and monitoring workflows.

Uber | Senior Engineer - AI Platform (Michelangelo)

MAY 2022 - FEB 2026

- Evolved the Michelangelo Feature Store into a pluggable, lightweight platform by adding external data-source plugin support without breaking existing user workflows.
- Designed, implemented, and released vector feature support in Palette; reduced CPU utilization from 200% to 125% and request latency from 2.25 seconds to 657 milliseconds in benchmark workloads.
- Drove an immutable Cassandra store initiative to improve production stability and deployment confidence across services.
- Built and launched a feature deprecation service and automated legacy-feature removal with validation scripts, enabling safe lifecycle management without impacting live ML models.
- Designed and deployed a shadow-testing environment for evaluating model performance on live production data in an isolated setting.
- Implemented E2E integration tests for near-real-time features, found and resolved a production bug, and strengthened deployment safety across Online Prediction Services.

Compass | Senior Machine Learning Engineer II (Tech Lead)

OCT 2020 - MAY 2022

- Led a team of 10+ software engineers and ML scientists building the Rekognition microservice for client-facing AI model inference.
- Designed the architecture for Document Intelligence, including document-processing, feature-extraction, and human-review services.
- Provided technical leadership for human-in-the-loop system architecture and AI-powered content-generation products.
- Managed design and development of video-generation systems built from 3D house models.

AiCure | Founding Engineer - Computer Vision & Machine Learning Platform

JUL 2012 - OCT 2020

- Designed and integrated a highly parallelized continuous-learning framework on AWS, enabling ML algorithms to improve as new data accumulated.
- Built an auto-annotation tool that reduced human effort in data labeling by 95%.
- Developed computer-vision and audio-analysis systems for object recognition, face recognition, sound detection, OCR, and behavior analysis.
- Built Android and iOS computer-vision applications that became core company products, including remote patient enrollment during the COVID-19 pandemic.

EDUCATION

Temple University | M.S. in Electrical and Computer Engineering

2009 - 2011

University of Science and Technology of China | B.S. in Automation

2006 - 2009

PATENTS

- Method and Apparatus for Recognition of Inhaler Actuation, US20130063579A1, filed Nov. 12, 2012, issued Mar. 14, 2013
- Apparatus and Method for Recognition of Suspicious Activities, US20140184772A1, filed Mar. 14, 2013, issued Jul. 3, 2014
- Apparatus and Method for Recognition of Medication Administration Indicator, US9317916B1, filed Apr. 22, 2013, issued Apr. 19, 2016
- Method and Apparatus for Recognition of Patient Activity, US20160196503A1, filed Jan. 6, 2015, issued Jul. 7 2016